Low-Power 16-bit ALU Design Utilizing Clock Gating and DVFS

Abstract

This project presents the design, verification, and physical implementation of a low-power 16-bit Arithmetic Logic Unit (ALU). To mitigate dynamic power dissipation, a hybrid power-saving methodology was employed, combining structural Clock Gating (CG) and Dynamic Voltage and Frequency Scaling (DVFS). The design was synthesized using Cadence Genus and progressed through a complete physical design flow in Cadence Innovus. Pre-layout analysis demonstrated a 46% reduction in dynamic power through clock gating alone. Further DVFS optimization reduced the theoretical power footprint to 38.19 μW. The final physical layout successfully achieved timing closure at the target 50 MHz frequency with a substantial setup slack of +14.815 ns. The layout achieved zero Design Rule Check (DRC) or Connectivity violations, resulting in a physically robust design with a final post-layout power consumption of 76.05 μW.

1. Introduction and Motivation

Power consumption is a primary bottleneck in modern Very Large Scale Integration (VLSI) systems. The Arithmetic Logic Unit (ALU) is one of the most active and power-hungry components within a microprocessor datapath. In standard ungated designs, clock signals continuously toggle all sequential elements, consuming dynamic switching power even when the ALU is idle.

The motivation of this project is to implement a 16-bit ALU that intelligently minimizes its power footprint. By introducing an `enable` signal to structurally gate the clock, dynamic switching is halted during idle clock cycles. Additionally, by applying analytical DVFS models, the operating voltage can be scaled down to the optimal threshold that still meets the target frequency, yielding a compounded power reduction.

2. Technical Specifications

- **Architecture:** 16-bit ALU supporting standard arithmetic, logical, and shift operations.
- **Technology Node:** Generic 45nm standard cell library (`slow_vdd1v0_basicCells`).
- **Nominal Supply Voltage (V_{DD}):** 1.0 V.
- **Target Frequency:** 50 MHz (20 ns clock period).
- **EDA Tools Used:** Cadence Xcelium (Simulation), Cadence Genus (Synthesis), Cadence Innovus (Place & Route).

3. Methodology

The project was executed through a standard four-phase ASIC design flow:

1. **RTL Design (Phase 1):** Two variants of the 16-bit ALU were developed in SystemVerilog. The *Baseline* variant unconditionally updates registers on every clock edge. The *Clock-Gated* variant uses an `enable` signal to freeze the operand registers and bypass the switching logic during idle states.

2. **Functional Verification (Phase 2):** A layered testbench was developed and simulated in Cadence Xcelium. Value Change Dump (VCD) files were generated to capture realistic switching activity, highlighting idle periods.

3. **Logic Synthesis & DVFS Calculation (Phase 3):** Both ALUs were synthesized in Cadence Genus using the `slow_vdd1v0` library. The VCD files were fed into Genus to perform activity-driven power estimation. A Python script was then used to extract the proportionality constant (K) and calculate the optimized voltage (V_{DDopt}) based on the critical path delay.

4. **Physical Design (Phase 4):** The synthesized netlist was imported into Cadence Innovus. The floorplan was sized at a 50% utilization target to accommodate Clock Tree Synthesis (CTS) buffers. Power planning, standard cell placement, CTS, and NanoRouting were performed to generate a fully DRC-clean GDSII layout.

4. Results and Analysis

4.1. Pre-Layout Synthesis Results

Synthesis was performed at a target frequency of 50 MHz using realistic simulation activity data. The integration of clock gating successfully reduced the dynamic switching power by nearly half.

- **Baseline ALU (Ungated):** Total Power: 74.16 μW | Critical Path Delay: 17.724 ns

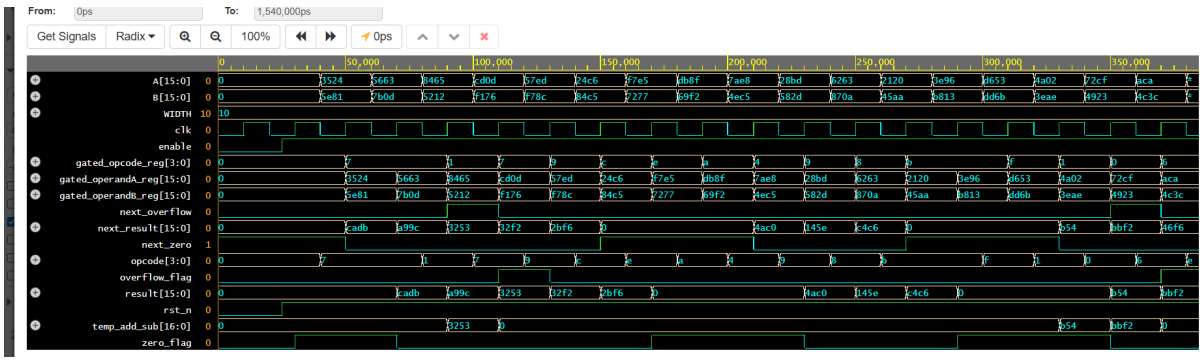


Figure 1: Non-Gated Clock Alu's Vector Waveform

- **Clock-Gated ALU:** Total Power: 39.99 μW | Critical Path Delay: 17.431 ns

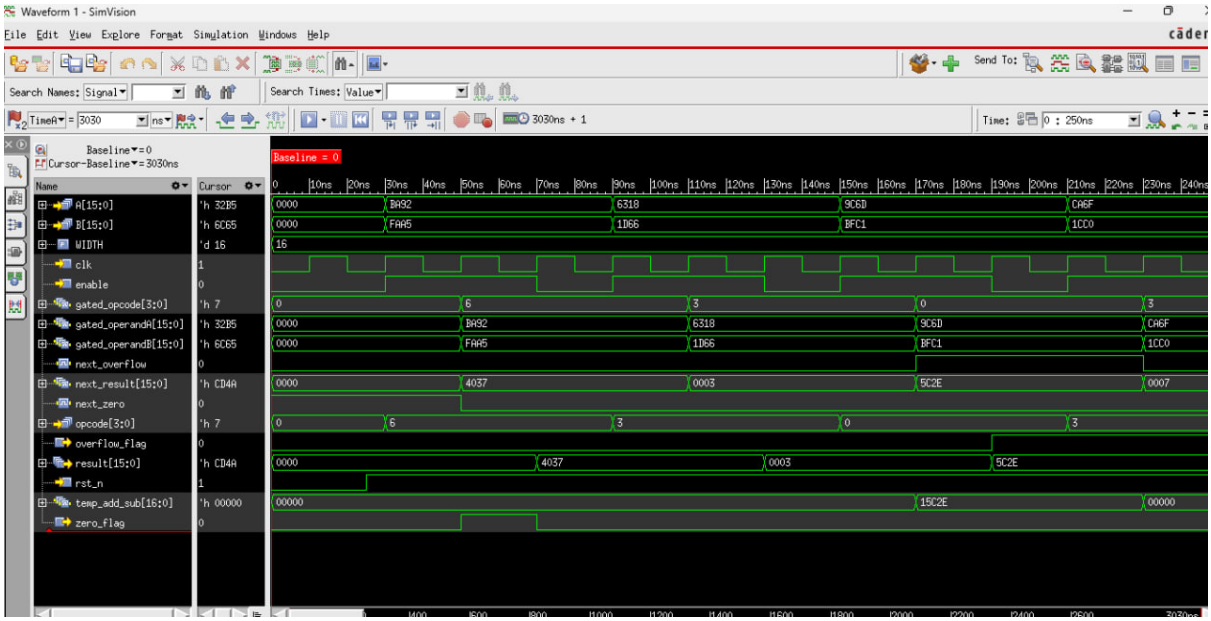


Figure 2: Gated Clock Alu's Vector Waveform

4.2. DVFS Optimization (Mode Table)

Using the parameters extracted from Genus ($C = 1.483$ pF, $f_{max} = 56.42$ MHz, $K = 3.545$ ns), the optimal supply voltage (V_{DDopt}) required to maintain the 50 MHz target was calculated to be 0.977 V. Combining DVFS with Clock Gating yields "Mode 2," the most power-efficient theoretical state.

Table 1: Calculated DVFS Operating Modes

Operating Mode	Description	Voltage (V)	Frequency (MHz)	Total Power (μW)
Mode 0	Baseline	1.000	50	74.16
Mode 1	DVFS Only	0.977	50	70.82
Mode 2	Hybrid Gated	0.977	50	**38.19**

4.3. Post-Layout Physical Design Results

The gated ALU design was successfully placed and routed in Cadence Innovus at the target 50 MHz (20 ns) clock constraint.

- **Timing Closure:** The physical layout easily met the 50 MHz setup constraint, achieving a massive positive Setup Slack of **+14.815 ns**, indicating the design is highly optimized and capable of operating at much higher frequencies.
- **Post-Layout Power:** Taking physical wire parasitics and the synthesized clock tree into account, the total post-layout power measured **76.05 μW** (0.076 mW).
- **Physical Verification:** The final NanoRouted design returned **0 Geometry (DRC) violations** and **0 Connectivity violations**, rendering it tape-out ready.

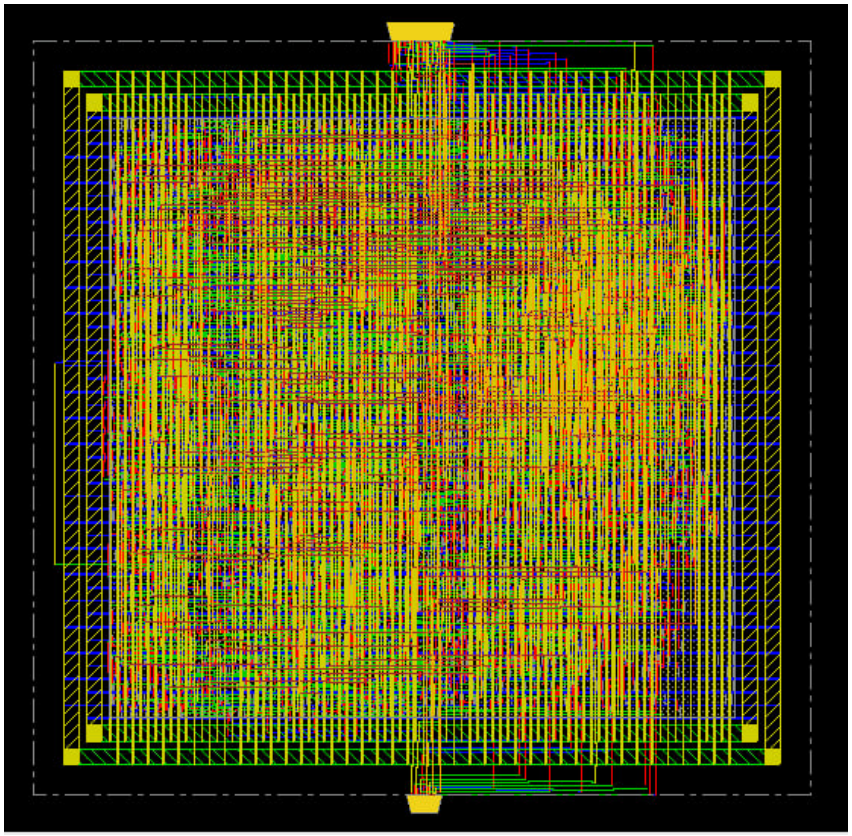
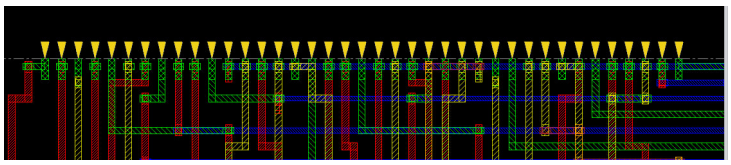
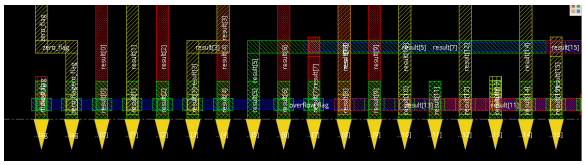


Figure 3: Gated Clock Alu's Innovus Physical Design .



(a) input pin



(b) output pin

Figure 4: Input and output pin layouts

4.4. GitHub Link

The generated reports,codes and implementation details for the 16-bit ALU are available on GitHub: Maruf2006191/16_bit-ALU

5. Challenges and Mitigations

- **Synthesis of Verification Constructs:** By default, Genus assumes a flat 20% switching activity for every single wire in the design.It assumes inputs are always

changing and clock is always triggering the registers. If we use vectorless estimation on a clock-gated design, our power results actually look worse than the baseline.Because adding clock gating requires inserting extra physical logic gates to control the clock.
Mitigation: We specify the `waves.vcd` file generated by simulating my testbench in genus `test.tcl` file, which tells genus the exact switching pattern of our RTL design .

- **Physical Design Congestion:** During the initial Innovus run with an 85% core utilization target, the insertion of CTS buffers pushed the density past 100%, causing 1,209 overlapping instances and over 1,000 DRC short-circuit violations.
Mitigation: The floorplan utilization was reduced to 50%, and the I/O boundary margins were expanded to 10 μm. This provided the NanoRouter ample physical space to isolate the power rings and cleanly route all metal wires without a single collision.

- **Post-Layout Power Overhead:** The final post-layout power (76.05 μW) exceeded the pre-layout prediction (39.99 μW).
Mitigation/Analysis: This increase is expected in a standard ASIC flow. Pre-layout analysis uses statistical wireload models. Post-layout analysis extracts exact RC parasitics from the physical metal routing and includes the additional dynamic power consumed by the inserted Clock Tree Synthesis (CTS) buffer cells, providing a realistic tape-out power figure.

6. Conclusion

This project successfully validated a low-power design methodology for a 16-bit ALU. By integrating structural clock gating, dynamic power was fundamentally reduced by eliminating unnecessary register toggling. The application of analytical DVFS modeling further minimized the theoretical power envelope to 38.19 μW. The transition into physical design proved the layout's structural integrity, comfortably achieving timing closure at 50 MHz (+14.815 ns slack) with zero DRC violations. The methodologies applied herein form a highly effective framework for designing power-conscious, high-performance datapath components.